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4.4 Answer Explanations

The following discussion is intended to familiarize you with the most efficient and effective approaches to the kinds of problems common to Problem-Solving questions. The particular questions in this chapter are generally representative of the kinds of problem-solving questions you will encounter on the GMAT exam. Remember that it is the problem-solving strategy that is important, not the specific details of a particular question.

Questions 1 to 82 — Difficulty: Easy

1. Working at a constant rate, a copy machine makes 20 copies of a one-page document per minute. If the machine works at this constant rate, how many hours does it take to make 4,800 copies of a one-page document?

   A. 4

   B. 5

   C. 6

   D. 7

   E. 8

   Arithmetic Rate

   The copy machine produces 20 copies of the one-page document each minute. Because there are 60 minutes in an hour, the constant rate of 20 copies per minute is equal to 60 × 20 = 1,200 copies per hour. With the machine working at this rate, the amount of time that it takes to produce 4,800 copies of the document is

   $$\frac{4800\ copies}{1200\ \frac{copies}{hour}} = 4\ \text{hours}.$$

   The correct answer is A.

2. If $x + y = 2$ and $x^2 + y^2 = 2$, what is the value of xy?

   A. –2

   B. –1

   C. 0

   D. 1

   E. 2

   Algebra Second-Degree Equations

$x + y = 2$	given
$y = 2 - x$	subtract x from both sides
$x^2 + (2 - x)^2 = 2$	substitute $y = 2 - x$ into $x^2 + y^2 = 2$
$2x^2 - 4x + 4 = 2$	expand and combine like terms
$2x^2 - 4x + 2 = 0$	subtract 2 from both sides
$x^2 - 2x + 1 = 0$	divide both sides by 2
$(x - 1)(x - 1) = 0$	factor
$x = 1$	set each factor equal to 0

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$y = 1$	use $x = 1$ and $y = 2 - x$
$xy = 1$	multiply 1 and 1

Alternatively, the value of xy can be found by first squaring both sides of the equation $x + y = 2$.

$x + y = 2$	given
$(x + y)^2 = 4$	square both sides
$x^2 + 2xy + y^2 = 4$	expand and combine like terms
$2 + 2xy = 4$	replace $x^2 + y^2$ with 2
$2xy = 2$	subtract 2 from both sides
$xy = 1$	divide both sides by 2

The correct answer is D.

3. The sum S of the first n consecutive positive even integers is given by $S = n(n + 1)$. For what value of n is this sum equal to 110?

   A. 10

   B. 11

   C. 12

   D. 13

   E. 14

   Algebra Factoring

   Given that the sum of the first n even numbers is $n(n + 1)$, the sum is equal to 110 when $110 = n(n + 1)$. To find the value of n in this case, we need to find the two consecutive integers whose product is 110. These integers are 10 and 11; 10 × 11 = 110. The smaller of these numbers is n.

   The correct answer is A.

4. 6(87.30 + 0.65) − 5(87.30) =

   A. 3.90

   B. 39.00

   C. 90.90

   D. 91.20

   E. 91.85

   Arithmetic Factors, Multiples, and Divisibility

   This question is most efficiently answered by distributing the 6 over 87.30 and 0.65, and then combining the terms that contain a factor of 87.30, as follows:

   6(87.30 + 0.65) − 5(87.30) = 6(87.30) + 6(0.65) − 5 (87.30) = (6 − 5) 87.30 + 6(0.65) = 87.30 + 3.90 = 91.20

   The correct answer is D.

5. What is the value of $x^2yz - xyz^2$, if $x = -2$, $y = 1$, and $z = 3$?

   A. 20

   B. 24

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C. 30

D. 32

E. 48

Algebra Operations on Integers

Given that $x = -2$, $y = 1$, and $z = 3$, it follows by substitution that

$$\begin{aligned} x^2yz - xyz^2 &= (-2)^2(1)(3) - (-2)(1)(3^2) \\ &= (4)(1)(3) - (-2)(1)(9) \\ &= 12 - (-18) \\ &= 12 + 18 \\ &= 30 \end{aligned}$$

The correct answer is C.

6. If T is a set of 35 consecutive integers, of which 17 are negative, what is the sum of all the integers in T?

A. –2

B. –1

C. 0

D. 1

E. 2

Arithmetic Properties of Numbers

The only set of 35 consecutive integers of which exactly 17 are negative is {−17, −16, −15, …, 0, …, 15, 16, 17}. The sum of these integers is (−17 + 17) + (−16 + 16) + (−15 + 15) + … + 0 = 0 + 0 + 0 + … + 0 = 0.

The correct answer is C.

7. If $x > y$ and $y > z$, which of the following represents the greatest number?

A. $x - z$

B. $x - y$

C. $y - x$

D. $z - y$

E. $z - x$

Algebra Inequalities

From $x > y$ and $y > z$, it follows that $x > z$. These inequalities imply the following about the differences that are given in the answer choices:

Answer choice	Difference	Algebraic sign	Reason
A	$x - z$	positive	$x > z$ implies $x - z > 0$
B	$x - y$	positive	$x > y$ implies $x - y > 0$
C	$y - x$	negative	$x - y > 0$ implies $y - x < 0$
D	$z - y$	negative	$y > z$ implies $0 > z - y$
E	$z - x$	negative	$x - z > 0$ implies $z - x < 0$

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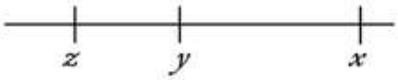
Since the expressions in A and B represent positive numbers and the expressions in C, D, and E represent negative numbers, the latter can be eliminated because every negative number is less than every positive number. To determine which of $x - z$ and $x - y$ is greater, consider the placement of points with coordinates x, y, and z on the number line.



The distance between x and z (that is, $x - z$) is the sum of the distance between x and y (that is, $x - y$) and the distance between y and z (that is, $y - z$). Therefore, $(x - z) > (x - y)$, which means that $x - z$ represents the greater of the numbers represented by $(x - z)$ and $(x - y)$. Thus, $x - z$ represents the greatest of the numbers represented by the answer choices.

Alternatively,

$y > z$	given
$-y < -z$	multiply both sides by –1
$x - y < x - z$	add x to both sides

Thus, $x - z$ represents the greater of the numbers represented by $(x - z)$ and $(x - y)$. Therefore, $x - z$ represents the greatest of the numbers represented by the answer choices.

The correct answer is A.

8. 38, 69, 22, 73, 31, 47, 13, 82

Which of the following numbers is greater than three-fourths of the numbers but less than one-fourth of the numbers in the list above?

A. 56

B. 68

C. 69

D. 71

E. 73

Arithmetic Order

Let's rearrange the eight numbers in the list from least to greatest: 13, 22, 31, 38, 47, 69, 73, 82. A number greater than three-fourths of them will be greater than each of the first six but less than either of the last two. Thus, this number must be greater than 69 but less than 73. Among the answer options, only the number 71 meets this condition.

The correct answer is D.

9. A rug manufacturer produces rugs at a cost of $75 per rug. What is the manufacturer's gross profit from the sale of 150 rugs if $\frac{2}{3}$ of the rugs are sold for $150 per rug and the rest are sold for $200 per rug?

A. $10,350

B. $11,250

C. $13,750

D. $16,250

E. $17,800

Arithmetic Applied Problems; Proportions

The gross profit from the sale of 150 rugs is equal to the revenue from the sale of the rugs minus the cost of

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The gross profit from the sale of 150 rugs is equal to the revenue from the sale of the rugs minus the cost of producing them. For $\frac{2}{3}$ of the 150 rugs—100 of them—the gross profit per rug is \$150 − \$75 = \$75. For the remaining 50 rugs, the gross profit per rug is \$200 − \$75 = \$125. The gross profit from the sale of the 150 rugs is therefore 100 × \$75 + 50 × \$125 = \$13,750.

The correct answer is C.

10. The value of Maureen's investment portfolio has decreased by 5.8 percent since her initial investment in the portfolio. If her initial investment was \$16,800, what is the current value of the portfolio?

    A. \$7,056.00

    B. \$14,280.00

    C. \$15,825.60

    D. \$16,702.56

    E. \$17,774.40

Arithmetic Percents

Maureen's initial investment was \$16,800, and it has decreased by 5.8%. Its current value is therefore (100% − 5.8%) = 94.2% of \$16,800, which is equal to 0.942 × \$16,800. To make the multiplication simpler, this can be expressed as \$(942 × 16.8). Thus multiplying, we obtain the result of \$15,825.60.

The correct answer is C.

11. Company C produces toy trucks at a cost of \$5.00 each for the first 100 trucks and \$3.50 for each additional truck. If 500 toy trucks were produced by Company C and sold for \$10.00 each, what was Company C's gross profit?

    A. \$2,250

    B. \$2,500

    C. \$3,100

    D. \$3,250

    E. \$3,500

Arithmetic Applied Problems

The company's gross profit on the 500 toy trucks is the company's revenue from selling the trucks minus the company's cost of producing the trucks. The revenue is (500)(\$10.00) = \$5,000. The cost for the first 100 trucks is (100)(\$5.00) = \$500, and the cost for the other 400 trucks is (400)(\$3.50) = \$1,400 for a total cost of \$500 + \$1,400 = \$1,900. Thus, the company's gross profit is \$5,000 – \$1,900 = \$3,100.

The correct answer is C.

	Profit or Loss (in millions of dollars)				
Division	1991	1992	1993	1994	1995
A	1.1	(3.4)	1.9	2.0	0.6
B	(2.3)	5.5	(4.5)	3.9	(2.9)
C	10.0	(6.6)	5.3	1.1	(3.0)

12. The annual profit or loss for the three divisions of Company T for the years 1991 through 1995 are summarized in the table shown, where losses are enclosed in parentheses. For which division and which three consecutive years shown was the division's profit or loss for the three-year period closest to \$0?

    A. Division A for 1991–1993

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B. Division A for 1992–1994

C. Division B for 1991–1993

D. Division B for 1993–1995

E. Division C for 1992–1994

Arithmetic Applied Problems

For completeness, the table shows all 9 of the profit or loss amounts, in millions of dollars, for each of the 3 divisions and the 3 three-year periods.

	1991–1993	1992–1994	1993–1995
A	–0.4	0.5	4.5
B	–1.3	4.9	–3.5
C	8.7	**–0.2**	3.4

The correct answer is E.

13. A computer appliance dealer bought a computer at a wholesale price of $500 and then sold it at a 20 percent discount off the suggested retail price. If the dealer made a 12 percent profit on the wholesale price, what was the suggested retail price of the computer?

A. $560

B. $580

C. $660

D. $700

E. $730

Arithmetic Applied Problems

The dealer's profit of 12 percent of the $500 wholesale price was 0.12 × $500 = $60. Thus, the dealer sold the computer for $500 + $60 = $560. Since this sale was at a 20 percent discount off the suggested retail price, $560 was 80 percent $\left(\frac{4}{5}\right)$ of the suggested retail price. Therefore, the suggested retail price was $\frac{\$560}{0.8} = \$700$.

The correct answer is D.

14. Pat invested x dollars in a fund that paid 8 percent annual interest, compounded annually. Which of the following represents the value, in dollars, of Pat's investment plus interest at the end of 5 years?

A. $5(0.08x)$

B. $5(1.08x)$

C. $[1 + 5(0.08)]x$

D. $(1.08)^5 x$

E. $(1.08x)^5$

Arithmetic Applied Problems

The formula for the compound interest earned on a principal x over n periods at percent rate r is $\left(1 + \frac{r}{100}\right)^n x - x$. For Pat's investment, that means the compound interest in dollars over 5 years is $(1.08)^5 x - x$. The total dollar value of Pat's investment includes the principal x plus the compound interest, so the total dollar value after 5 years is $((1.08)^5 x - x) + x = (1.08)^5 x$.

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The correct answer is D.

15. There are five sales agents in a certain real estate office. One month Andy sold twice as many properties as Ellen, Bob sold 3 more than Ellen, Cary sold twice as many as Bob, and Dora sold as many as Bob and Ellen together. Who sold the most properties that month?

    A. Andy

    B. Bob

    C. Cary

    D. Dora

    E. Ellen

Algebra Order

Let x represent the number of properties that Ellen sold, where $x \geq 0$. Then, since Andy sold twice as many properties as Ellen, $2x$ represents the number of properties that Andy sold. Bob sold 3 more properties than Ellen, so $(x+3)$ represents the number of properties that Bob sold. Cary sold twice as many properties as Bob, so $2(x+3) = (2x+6)$ represents the number of properties that Cary sold. Finally, Dora sold as many properties as Bob and Ellen combined, so $[(x+3)+x] = (2x+3)$ represents the number of properties that Dora sold. The following table summarizes these results.

Agent	Properties Sold
Andy	$2x$
Bob	$x+3$
Cary	$2x+6$
Dora	$2x+3$
Ellen	x

Since $x \geq 0$, clearly $2x+6$ exceeds x, $x+3$, $2x$, and $2x+3$. Therefore, Cary sold the most properties.

The correct answer is C.

16. In a field day at a school, each child who competed in n events and scored a total of p points was given an overall score of $\frac{p}{n} + n$. Andrew competed in 1 event and scored 9 points. Jason competed in 3 events and scored 5, 6, and 7 points, respectively. What was the ratio of Andrew's overall score to Jason's overall score?

    A. $\frac{10}{23}$

    B. $\frac{7}{10}$

    C. $\frac{4}{5}$

    D. $\frac{10}{9}$

    E. $\frac{12}{7}$

Algebra Applied Problems; Substitution

Andrew participated in 1 event and scored 9 points, so his overall score was $\frac{9}{1} + 1 = 10$. Jason participated in

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Andrew participated in 1 event and scored 9 points, so his overall score was $\frac{9}{1} + 1 = 10$. Jason participated in 3 events and scored 5 + 6 + 7 = 18 points, so his overall score was $\frac{18}{3} + 3 = 9$. The ratio of Andrew's overall score to Jason's overall score was $\frac{10}{9}$.

The correct answer is D.

17. A certain work plan for September requires that a work team, working every day, produce an average of 200 items per day. For the first half of the month, the team produced an average of 150 items per day. How many items per day must the team average during the second half of the month if it is to attain the average daily production rate required by the work plan?

    A. 225

    B. 250

    C. 275

    D. 300

    E. 350

    Arithmetic Rate Problem

    The work plan requires that the team produce an average of 200 items per day in September. Because the team has only produced an average of 150 items per day in the first half of September, it has a shortfall of 200 − 150 = 50 items per day for the first half of the month. The team must make up for this shortfall in the second half of the month, which has an equal number of days as the first half of the month. The team must therefore produce in the second half of the month an average amount per day that is 50 items greater than the required average of 200 items per day for the entire month. This amount for the second half of September is 250 items per day.

    The correct answer is B.

18. A company sells radios for $15.00 each. It costs the company $14.00 per radio to produce 1,000 radios and $13.50 per radio to produce 2,000 radios. How much greater will the company's gross profit be from the production and sale of 2,000 radios than from the production and sale of 1,000 radios?

    A. $500

    B. $1,000

    C. $1,500

    D. $2,000

    E. $2,500

    Arithmetic Applied Problems

    If the company produces and sells 1,000 radios, its gross profit from the sale of these radios is equal to the total revenue from the sale of these radios minus the total cost. The total cost is equal to the number of radios produced multiplied by the production cost per radio: 1,000 × $15.00. The total revenue is equal to the number of radios sold multiplied by the selling price: 1,000 × $14.00. The gross profit in this case is therefore 1,000 × $15.00 − 1,000 × $14.00 = 1,000 × ($15.00 − $14.00) = 1,000 ($1.00) = $1,000. If 2,000 radios are produced and sold, the total cost is equal to 2,000 × $13.50 and the total revenue is equal to 2,000 × $15.00. The gross profit in this case is therefore 2,000 × $15.00 − 2,000 × $13.50 = 2,000 × ($15.00 − $13.50) = 2,000 × ($1.50) = $3,000. This profit of $3,000 is $2,000 greater than the gross profit of $1,000 from producing and selling 1,000 radios.

    The correct answer is D.

19. Working at their respective constant rates, machine A makes 100 copies in 12 minutes and machine B makes 150 copies in 10 minutes. If these machines work simultaneously at their respective rates for 30 minutes, what is the total number of copies that they will produce?

    A. 250

    B. 425

    C. 675

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C. 675

D. 700

E. 750

Arithmetic Applied Problems

Over 30 minutes, machine A makes $100 \text{ copies} \times \left(\frac{30 \text{ minutes}}{12 \text{ minutes}}\right) = 250$ copies, while machine B makes $150 \text{ copies} \times \left(\frac{30 \text{ minutes}}{10 \text{ minutes}}\right) = 450 \text{ copies}$. So together, the machines will make 250 + 450 = 700 copies in 30 minutes.

The correct answer is D.

20. Company X mailed out a questionnaire. Sixty percent of those who received the questionnaire by mail responded. Company X needed 300 responses. What is the minimum number of questionnaires that Company X should have mailed in order to get the necessary responses?

A. 400

B. 420

C. 480

D. 500

E. 600

Arithmetic Percents

Since 300, the number of responses received, was 60 percent of the number of copies of the questionnaire that were mailed, the number of copies mailed was $\frac{300}{0.6} = 500$.

The correct answer is D.

21. The toll T, in dollars, for a truck using a certain bridge is given by the formula $T = 1.50 + 0.50(x - 2)$, where x is the number of axles on the truck. What is the toll for an 18-wheel truck that has 2 wheels on its front axle and 4 wheels on each of its other axles?

A. $2.50

B. $3.00

C. $3.50

D. $4.00

E. $5.00

Algebra Operations on Rational Numbers

The 18-wheel truck has 2 wheels on its front axle and 4 wheels on each of its other axles, and so if A represents the number of axles on the truck in addition to the front axle, then $2 + 4A = 18$, from which it follows that $4A = 16$ and $A = 4$. Therefore, the total number of axles on the truck is $1 + A = 1 + 4 = 5$. Then, using $T = 1.50 + 0.50(x - 2)$, where x is the number of axles on the truck and $x = 5$, it follows that $T = 1.50 + 0.50(5 - 2) = 1.50 + 1.50 = 3.00$. Therefore, the toll for the truck is $3.00.

The correct answer is B.

22. For what value of x between −4 and 4, inclusive, is the value of $x^2 - 10x + 16$ the greatest?

A. −4

B. −2

C. 0

D. 2

E. 4

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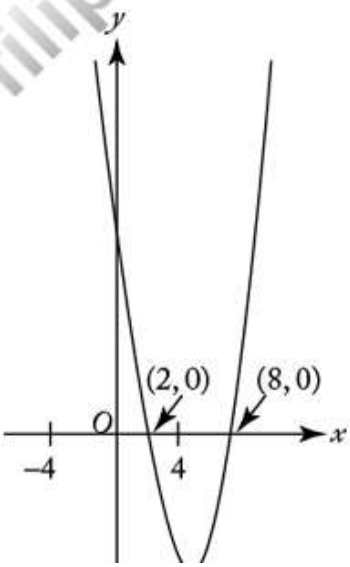
Algebra Second-Degree Equations

Given the expression $x^2 - 10x + 16$, a table of values can be created for the corresponding function $f(x) = x^2 - 10x + 16$ and the graph in the standard (x,y) coordinate plane can be sketched by plotting selected points:

x	$f(x)$
−4	72
−3	55
−2	40
−1	27
0	16
1	7
2	0
3	−5
4	−8
5	−9
6	−8
7	−5
8	0
9	7



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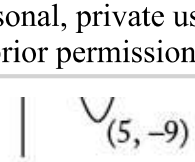

It is clear from both the table of values and the sketch of the graph that as the value of x increases from −4 to 4, the values of $x^2 - 10x + 16$ decrease. Therefore, the value of $x^2 - 10x + 16$ is greatest when $x = -4$.

Alternatively, the given expression, $x^2 - 10x + 16$, has the form $ax^2 + bx + c$, where $a = 1$, $b = -10$, and $c = 16$. The graph in the standard (x,y) coordinate plane of the corresponding function $f(x) = ax^2 + bx + c$ is a parabola with vertex at $x = -\frac{b}{2a}$, and so the vertex of the graph of $f(x) = x^2 - 10x + 16$ is at

$x = -\left(\frac{-10}{2(1)}\right) = 5.$

Because $a = 1$ and 1 is positive, this parabola opens upward and values of $x^2 - 10x + 16$ decrease as x increases from −4 to 4. Therefore, the greatest value of $x^2 - 10x + 16$ for all values of x between −4 and 4, inclusive, is at $x = -4$.

The correct answer is A.

23. If $x = -\frac{5}{8}$ and $y = -\frac{1}{2}$, what is the value of the expression $-2x - y^2$?

A. $-\frac{3}{2}$

B. -1

C. 1

D. $\frac{3}{2}$

E. $\frac{7}{4}$

Algebra Fractions

If $x = -\frac{5}{8}$ and $y = -\frac{1}{2}$, then $-2x - y^2 = -2\left(-\frac{5}{8}\right) - \left(-\frac{1}{2}\right)^2 = \frac{5}{4} - \frac{1}{4} = \frac{4}{4} = 1.$

The correct answer is C.

24. At the opening of a trading day, stock X had a value of \$40.00 per share and stock Y had a value of \$25.00 per share. At the closing of the trading day, stock X had a value of \$41.00 per share and stock Y had a value of \$28.50 per share. For this trading day, the percent increase in the value of stock Y was how many percentage points greater than the percent increase in the value of stock X?

A. 2.5

B. 6.25

C. 10

D. 11.5

E. 12.5

Arithmetic Percents

Stock X's percentage increase in value was $(41 - 40) \times \frac{100\%}{40} = \frac{100\%}{40} = 2.5\%$. Stock Y's was $(28.5 - 25) \times \frac{100\%}{25} = \frac{350\%}{25} = 14\%$. Thus, the percentage increase in stock Y's value was (14 − 2.5)% = 11.5 percentage points greater than that in stock X's value.

The correct answer is D.

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25. For positive integers a and b, the remainder when a is divided by b is equal to the remainder when b is divided by a. Which of the following could be a value of ab?

    I. 24

    II. 30

    III. 36

    A. II only

    B. III only

    C. I and II only

    D. II and III only

    E. I, II, and III

    Arithmetic Properties of Integers

    We are given that the remainder when a is divided by b is equal to the remainder when b is divided by a, and asked about possible values of ab. We thus need to find what our given condition implies about a and b.

    We consider two cases: $a = b$ and $a \neq b$.

    If $a = b$, then our given condition is trivially satisfied: the remainder when a is divided by a is equal to the remainder when a is divided by b. The condition thus allows that a be equal to b.

    Now consider the case of $a \neq b$. Either a < b or $b < a$. Supposing that $a < b$, the remainder when a is divided by b is simply a. (For example, if 7 is divided by 10, then the remainder is 7.) However, according to our given condition, this remainder, a, is also the remainder when b is divided by a, which is impossible. If b is divided by a, then the remainder must be less than a. (For example, for any number that is divided by 10, the remainder cannot be 10 or greater.) Similar reasoning applies if we suppose that $b < a$. This is also impossible.

    We thus see that a must be equal to b, and consider the statements I, II, and III.

    I. Factored in terms of prime numbers, 24 = 3 × 2 × 2 × 2. Because "3" occurs only once in the factorization, we see that there is no integer a such that $a \times a = 24$. Based on the reasoning above, we see that 24 cannot be a value of ab.

    II. Factored in terms of prime numbers, 30 = 5 × 3 × 2. Because there is no integer a such that $a \times a = 30$, we see that 30 cannot be a value of ab.

    III. Because 36 = 6 × 6, we see that 36 is a possible value of ab (with $a = b$).

    The correct answer is B.

26. List S consists of the positive integers that are multiples of 9 and are less than 100. What is the median of the integers in S?

    A. 36

    B. 45

    C. 49

    D. 54

    E. 63

    Arithmetic Series and Sequences

    In the set of positive integers less than 100, the greatest multiple of 9 is 99 (9 × 11) and the least multiple of 9 is 9 (9 × 1). The sequence of positive multiples of 9 that are less than 100 is therefore the sequence of numbers $9 \times k$, where k ranges from 1 through 11. The median of the numbers k from 1 through 11 is 6. Therefore, the median of the numbers $9 \times k$, where k ranges from 1 through 11, is 9 × 6 = 54.

    The correct answer is D.

27. A rope 20.6 meters long is cut into two pieces. If the length of one piece of rope is 2.8 meters shorter than the length of the other, what is the length, in meters, of the longer piece of rope?

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A. 7.5

B. 8.9

C. 9.9

D. 10.3

E. 11.7

Algebra First-Degree Equations

If x represents the length of the longer piece of rope, then $x - 2.8$ represents the length of the shorter piece, where both lengths are in meters. The total length of the two pieces of rope is 20.6 meters so,

$$\begin{aligned} x + (x - 2.8) &= 20.6 && \text{given} \\ 2x - 2.8 &= 20.6 && \text{add like terms} \\ 2x &= 23.4 && \text{add 2.8 to both sides} \\ x &= 11.7 && \text{divide both sides by 2} \end{aligned}$$

Thus, the length of the longer piece of rope is 11.7 meters.

The correct answer is E.

28. If x and y are integers and $x - y$ is odd, which of the following must be true?

I. xy is even.

II. $x^2 + y^2$ is odd.

III. $(x + y)^2$ is even.

A. I only

B. II only

C. III only

D. I and II only

E. I, II, and III

Arithmetic Properties of Numbers

We are given that x and y are integers and that $x - y$ is odd, and then asked, for various operations on x and y, whether the results of the operations are odd or even. It is therefore useful to determine, given that $x - y$ is odd, whether x and y are odd or even. If both x and y are even—that is, divisible by 2—then $x - y = 2m - 2n = 2(m - n)$ for integers m and n. We thus see if both x and y are even then $x - y$ cannot be odd. And because $x - y$ *is* odd, we see that x and y cannot both be even. Similarly, if both x and y are odd, then, for integers j and k, $x = 2j + 1$ and $y = 2k + 1$. Therefore, $x - y = (2j + 1) - (2k + 1)$. The ones cancel, and we are left with $x - y = 2j - 2k = 2(j - k)$. Because $2(j - k)$ would be even, x and y cannot both be odd if $x - y$ is odd. It follows from all of this that one of x or y must be even and the other odd.

Now consider the statements I through III.

I. If one of x or y is even, then one of x or y is divisible by 2. It follows that xy is divisible by 2 and that xy is even.

II. Given that a number x or y is odd—not divisible by 2—we know that its product with itself is not divisible by 2 and is therefore odd. On the other hand, given that a number x or y is even, we know that its product with itself *is* divisible by 2 and is therefore even. The sum $x^2 + y^2$ is therefore the sum of an even number and an odd number. In such a case, the sum can be written as $(2m) + (2n + 1) = 2(m + n) + 1$, with m and n integers. It follows that $x^2 + y^2$ is not divisible by 2 and is therefore odd.

III. We know that one of x or y is even and the other is odd. We can therefore see from the discussion of

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III. We know that one of x or y is even and the other is odd. We can therefore see from the discussion of statement II that $x + y$ is odd, and then also see, from the discussion of statement II, that the product of $x + y$ with itself, $(x + y)^2$, is odd.

The correct answer is D.

29. On Monday, the opening price of a certain stock was $100 per share and its closing price was $110 per share. On Tuesday the closing price of the stock was 10 percent less than its closing price on Monday, and on Wednesday the closing price of the stock was 4 percent greater than its closing price on Tuesday. What was the approximate percent change in the price of the stock from its opening price on Monday to its closing price on Wednesday?

A. A decrease of 6%

B. A decrease of 4%

C. A decrease of 1%

D. An increase of 3%

E. An increase of 4%

Arithmetic Percents

The closing share price on Tuesday was 10% less than the closing price on Monday, $110. 10% of $110 is equal to 0.1 × $110 = $11, so the closing price on Tuesday was $110 − $11 = $99. The closing price on Wednesday was 4% greater than this: $99 + (0.04 × $99) = $99 + $3.96 = $102.96. This value, $102.96, is 2.96% greater than $100, the opening price on Monday. The percentage change from the opening share price on Monday is therefore an increase of approximately 3%, which is the closest of the available answers to an increase of 2.96%.

The correct answer is D.

30. 1 − 0.000001 =

A. (1.01) (0.99)

B. (1.11) (0.99)

C. (1.001) (0.999)

D. (1.111) (0.999)

E. (1.0101) (0.0909)

Arithmetic Place Value

The task in this question is to find among the available answers the expression that is equal to 1 − 0.000001 = 0.999999. In the case of answer choice C, the first of the two factors, (1.001), is equal to 1 + 0.001. One may therefore observe that (1.001)(0.999) = (1 + 0.001)(0.999) = 0.999 + 0.000999 = 0.999999. Answer choice C is therefore a correct answer.

For answer choice A, (1.01)(0.99) = (1 + 0.01)(0.99) = 0.9999. This answer choice is therefore incorrect. For answer choice B, (1.11)(0.99) = (1 + 0.1 + 0.01)(0.99) = 0.99 + 0.099 + 0.0099 = 1.0989. This answer choice is therefore incorrect. For answer choice D, (1.111)(0.999) = 0.999 + 0.0999 + 0.00999 + 0.000999 = 1.109889. This answer choice is therefore incorrect. For answer choice E, (1.0101)(0.909) = 0.909 + 0.00909 + 0.0000909 = 0.9181809. This answer choice is therefore incorrect.

The correct answer is C.

31. In a certain history class of 17 juniors and seniors, each junior has written 2 book reports and each senior has written 3 book reports. If the 17 students have written a total of 44 book reports, how many juniors are in the class?

A. 7

B. 8

C. 9

D. 10

E. 11

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